

Zachary Gilliam

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Full-stack engineer targeting a role where software and healthcare meet. I've shipped production React/TypeScript and Node.js platforms in clinical environments that others only prototype, and I bring the full-stack ownership, CI/CD discipline, and cross-functional collaboration to help organizations build faster and more reliably.

Education



University of Iowa

Bachelor's in Arts, Human Computer Interaction

IP May 2026

Presidents list 3/8 Semesters Deans List 4/8 Semesters

[GPA: 3.56/4.00]

Master's of Science, Health Informatics

IP

Focus on Agile Product Development in Health-Tech Space

[GPA: 4.00/4.00]

Experience



Health Brain Cognition Lab

Aug 2024 - Present

Lead Developer — Fullstack // Data Platform

Python, PyTorch, SQL (MariaDB), CI/CD, Typescript, Linux, Bash, HPC

- **Designed and maintained automated data ingestion and quality pipelines** across 9+ large-scale datasets, standardizing multi-domain data (neuroimaging, genetics, etc.) for use by 30+ researchers across 4 active studies.
- Developed **agentic AI tooling** on top of existing data infrastructure, enabling natural-language querying, automated QA, and report generation — reducing turnaround time by 40% for non-technical team members.
- **Led a development team**, operationalizing CI/CD, structured PR governance, and engineering standards across 8 codebases, lowering production defects and increasing deployment frequency by ~60%.
- Developed a **production full-stack reporting platform** (TypeScript/React + Bayesian ML models) generating personalized analytics for thousands of users in large-scale intervention studies.
- Participated in various symposia, **winning two first place awards** at the UI computational psychiatry symposium (2024)



Cerebra // Clinical & Translational Science

Oct 2024 - Aug 2025

Project Lead - Fullstack Developer

Typescript (React), PostgreSQL (Supabase), Python, Pytorch

- Architected and shipped a **full-stack patient-provider communication platform** in a HIPAA-sensitive clinical environment, handling end-to-end delivery from database schema to production deployment.
- Integrated **Bayesian ML models** into the application backend to surface **real-time patient risk scores** and predicted PT recovery timelines, translating clinical data into actionable provider-facing UI.
- Built a **React/TypeScript frontend** with **real-time data visualization** and **secure messaging**, enabling providers to monitor patient progress and communicate within a single, unified interface.



More Experiences



Judgment, Decision, and Social Comparison Lab

Aug 2025 - Dec 2025

Lead Developer

TypeScript, Node, AWS, DynamoDB/PartiQL

- **Translated user requirements** into a **rapidly prototyped** TypeScript web application, delivering a lightweight decision-support tool used by thousands of users.
- **Built, tested, and deployed a cloud-native application** on AWS infrastructure, ensuring **reliable global access and low-latency performance** for thousands of concurrent users.
- **Translated stakeholder feedback into iterative improvements**, ensuring usability and research accuracy.



WorkHorseSCS

May 2022 - Aug 2024

Intern Developer / Process Automation Engineer

React, JS, HTML/CSS, SQL, Linux, AWS, Python

- Utilized **NLP to analyze user feedback** and generate help pages with AI, reducing user help ticket volume by 30%.
- Designed and implemented Node.js backend services and scalable data APIs **powering modular user-facing dashboards** built with JavaScript and SQL.
- Automated onboarding process, reducing employee involvement by 60%.

Skills

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Python

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Linux

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SQL

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AI integration

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Cloud Compute

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Agile Product Development

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Typescript

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HTML/CSS

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Git